PERFORMANCE TESTING REPORT

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance testing evaluates how your system behaves under expected and peak load conditions[cite: 26]. Complete the sections below to document your testing approach, tools used, and results obtained[cite: 26].

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter[cite: 26]
Type of Testing	Load Testing, Stress Testing, Spike Testing[cite: 26]
Target Module / API	Crop Recommendation and Optimization API (`/predict`)[cite: 26]
Test Environment	Localhost (Python Flask Server on Local Machine)[cite: 26]
Test Date	03 July 2026[cite: 26]

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	**Normal Load Test** Multiple users accessing prediction API[cite: 26]	10[cite: 26]	60[cite: 26]	System should respond within acceptable time with 100% success[cite: 26].
2	**High Load Test** Simulate heavy concurrent users[cite: 26]	50[cite: 26]	120[cite: 26]	System should handle load with response time < 2 sec and high success rate[cite: 26].
3	**Stress Test** Push system beyond normal capacity[cite: 26]	100[cite: 26]	180[cite: 26]	System should remain stable and not crash; graceful handling of requests[cite: 26].
4	**Spike Test** Sudden increase in user load[cite: 26]	200[cite: 26]	60[cite: 26]	System should handle spikes with minimal performance degradation[cite: 26].

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	**Response Time (Avg)**	< 2 seconds[cite: 26]	1.18 seconds[cite: 26]	**Pass[cite: 26]**	Average response time is within acceptable limit.[cite: 26]

S.No	Metric	Target Value	Actual Value	Status	Remarks
2	**Response Time (Max)**	< 5 seconds[cite: 26]	2.86 seconds[cite: 26]	**Pass[cite: 26]**	Maximum response time is within acceptable limit.[cite: 26]
3	**Throughput (Req/sec)**	-	148.7 req/sec[cite: 26]	**Pass[cite: 26]**	System handled high request rate successfully.[cite: 26]
4	**Error Rate**	-	0.32%[cite: 26]	**Pass[cite: 26]**	Error rate is less than 1%.[cite: 26]
5	**CPU Utilization**	-	63%[cite: 26]	**Pass[cite: 26]**	CPU usage is under 80% during test.[cite: 26]
6	**Memory Utilization**	-	56%[cite: 26]	**Pass[cite: 26]**	Memory usage is under 80% during test. [cite: 26]

Step 4: Observations & Analysis

Key Findings:

- The system responded within the acceptable time range for both average and maximum response time[cite: 26].
- Throughput was stable and the system handled concurrent user load efficiently[cite: 26].
- Error rate remained below 1% indicating reliable system behavior[cite: 26].

Bottlenecks Identified:

- High database query time during peak concurrent requests from regional farming entries[cite: 26].
- Some delay observed in loading dashboard analytics due to heavy multi-variable agricultural data processing[cite: 26].

Optimization Steps Taken:

- Optimized SQL queries and added indexing on frequently accessed soil parameter tables[cite: 26].
- Implemented caching for dashboard data to reduce response time for crop profiles[cite: 26].
- Enabled connection pooling to improve database transaction performance under load[cite: 26].

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):[cite: 26]

[Insert Screenshot 1 Here - JMeter Summary Report for Crop Prediction Core Module][cite: 26]

[Insert Screenshot 2 Here - Response Time Graph for Concurrent Soil Parameters Ingestion][cite: 26]